

Module 8: Hardware Integration, Handheld QR Devices, ESC/POS Printers & Server Setup Infrastructure

Comprehensive 12-week milestone agreement, engineering effort breakdown, resource cost schedule, and acceptance criteria based on the verified Business Analysis Report **ARCH-001 v1.0** for **Popular Auto Spare & A/C Parts Co. W.L.L.**

DOCUMENT ID

SL-POP-ERP-MS-008

TOTAL ENGINEERING EFFORT

3 Working Weeks (0.75 Months)

TARGET ARCHITECTURE

Multi-Branch Admin & Asset Tracking

BA REFERENCE DOCUMENT

ARCH-001 v1.0 (04-July-2026)

FIXED PROJECT COST

BD 4,570.909

DOCUMENT STATUS

FINAL PROPOSAL

Executive Summary & Master ERP Context

The **Enterprise Administration Modules** (Module 8) provide the operational infrastructure, asset tracking, facility governance, fleet control, and statutory legal document compliance framework for Popular Auto Spare & A/C Parts Co. W.L.L. As the enterprise expands across regional distribution hubs, retail branches, and central warehouses, managing physical facilities, tracking fixed assets with QR codes, optimizing fleet operations, and safeguarding legal corporate records become vital for cost control, security, and statutory readiness.



Core Objective & Business Purpose

To establish an integrated administrative operations platform consolidating Workplace Facility Management, QR-based Fixed Asset Lifecycle Tracking & Depreciation, Commercial Fleet & Vehicle Logistics Management, Staff Physical Locker Allocations, and Secure Digital Document & Statutory Renewal Governance.

Master ERP Transformation Architecture (9 Core Process Modules)

CURRENT SCOPE: MODULE 06
ACTIVE

- 01 PCode Gen & Item Master
- 02 Vendor & Purchase Flow
- 03 Stock Verification & Location
- 04 Sales & POS Checkout
- 05 Finance, Tax & VAT Accounting
- 06 Administration & Security
- 07 HRMS & Biometric Payroll

08 Hardware & Barcode Infrastructure

09 Executive BI & Mobile Analytics



Facility, Lease & Utility Control

Centralized facility registry, preventive/corrective work orders, utility expense tracking, property lease renewal alerts, and visitor access control.

Facility Master

Lease Alerts

Utility Log



QR Fixed Asset Lifecycle & Audit

QR code tagging, custodian assignment, multi-branch asset transfers, straight-line depreciation calculation, and annual physical audit verification.

QR Tagging

Depreciation

Asset Audit



Fleet Logistics & Document Vault

Commercial vehicle tracking, driver bookings, fuel consumption logs, insurance/road tax compliance, physical locker allocation, and corporate digital document repository.

Fleet TCO

Fuel Log

Digital Vault

Module 8: Functional Scope & Architecture

Breakdown (BA Ref: ARCH-001)

Derived directly from the 25 pages of **ARCH-001: Administration Modules Business Analysis Report**, the implementation scope is structured across 4 core operational pillars:

2.1 Facility & Workplace Office Management

- **Facility Master:** Centralized registry of corporate offices, regional branches, central warehouses, workshops, retail outlets, and staff accommodations.
- **Maintenance Work Orders:** Preventive scheduled maintenance calendars, AMC vendor contracts, breakdown ticketing, and repair sign-off.
- **Employee Service Requests:** Internal ticketing portal for electrical, HVAC/air conditioning, plumbing, carpentry, and IT infrastructure requests.
- **Utility & Consumption Control:** Monthly tracking of electricity, water, internet, telephone, gas, and generator fuel with budget variance analysis.
- **Visitor Management (VMS):** Digital visitor registration, badge generation, host employee notification, and entry/exit timestamp logging.
- **Lease & Property Governance:** Landlord agreements, rent payment schedules, commercial lease terms, and automated renewal notifications.

2.2 Fixed Asset Management & QR Lifecycle Tracking

- **Asset Master Registry:** Comprehensive cataloging of IT equipment, warehouse machinery, tools, office furniture, vehicles, and facilities.
- **QR Code Tagging & Scanner:** Unique asset QR label generation with handheld scanner integration for rapid location identification and audits.
- **Asset Custody & Movements:** Tracking of employee custodians, department assignments, and inter-branch transfer delivery challans.
- **Warranty & AMC Management:** Supplier warranty coverage tracking, AMC renewal alerts, and maintenance history logging.
- **Depreciation Engine:** Automated straight-line and reducing-balance depreciation posting integrated directly with General Ledger (GL).

- **Physical Audit & Disposal:** Periodic physical inventory verification, variance logging, asset write-off, and scrap disposal workflows.

2.3 Fleet Logistics, Vehicle Maintenance & Total Cost of Ownership

- **Vehicle Master Registry:** Profile management for commercial delivery vans, heavy transport trucks, pickups, and corporate cars.
- **Vehicle Allocation & Booking:** Driver assignment, delivery route scheduling, key management, and trip odometer logging.
- **GPS & Route Monitoring:** Integration with telematics for real-time delivery tracking and mileage verification.
- **Preventive Maintenance:** Service intervals by mileage/date, oil changes, tire replacement logs, and workshop repair history.
- **Fuel Management:** Fuel card reconciliation, fuel receipt uploads, kilometer-per-liter efficiency analysis, and fuel theft anomaly detection.
- **Statutory Vehicle Compliance:** Traffic registration renewals, comprehensive insurance policies, fitness certificates, and accident claim workflows.
- **Vehicle TCO Analytics:** Real-time Total Cost of Ownership (TCO) tracking per vehicle combining fuel, maintenance, insurance, and depreciation.

2.4 Staff Locker & Corporate Document Management System

- **Staff Locker Management:** Branch physical locker registry, employee key allocation, locker inspections, and surrender protocols.
- **Digital Document Repository:** Encrypted digital vault for Commercial Registrations (CR), municipality licenses, lease agreements, vehicle titles, and contracts.
- **Role-Based Access Control (RBAC):** Granular permissions restricting document access to authorized executive and HR/Admin personnel.
- **Physical Document Movement:** Barcoded checkout/check-in tracking for original legal deeds, bank guarantees, and official certificates.
- **Automated Expiry & Renewal Alerts:** Proactive 90/60/30-day notifications for CR renewals, chamber of commerce, civil defense, and trade permits.
- **Compliance Audit Readiness:** Full historical audit trail of document uploads, version updates, downloads, and custodian modifications.

Project Team Structure & Dedicated Resource Pricing (12 Weeks / 3.0 Months)

In strict accordance with the **SaNDS Lab Dedicated Engineering Rate Card**, the resource allocation and cost breakdown for the 12-week (3.0 months / 60 working days) delivery of Module 8 is structured as follows (all figures in Bahraini Dinars - BHD):

ENGINEERING ROLE	MONTHLY RATE (BHD)	WEEKLY RATE (BHD)	HOURLY RATE (160H/MO)	EFFORT ALLOCATION	TOTAL COST (BHD)
Project Manager & Enterprise Solutions Architect	BD 545.455	BD 136.364	BD 3.409	3.0 Months (12 Wks)	BD 1,636.365
Back-End Lead Engineer (PHP MVC / REST APIs)	BD 227.273	BD 56.818	BD 1.420	3.0 Months (12 Wks)	BD 681.819
Front-End Lead Engineer (Admin UI / React Dashboards)	BD 227.273	BD 56.818	BD 1.420	3.0 Months (12 Wks)	BD 681.819
Database & Cloud Infrastructure Architect	BD 204.545	BD 51.136	BD 1.278	3.0 Months (12 Wks)	BD 613.635
QA & System Test Automation	BD 159.091	BD 39.773	BD 0.994	3.0 Months (12 Wks)	BD 477.273

ENGINEERING ROLE	MONTHLY RATE (BHD)	WEEKLY RATE (BHD)	HOURLY RATE (160H/MO)	EFFORT ALLOCATION	TOTAL COST (BHD)
Lead Engineer					
Total Dedicated Engineering Team	BD 1,523.636 / mo	BD 380.909 / wk	BD 8.523 / hr	12 Weeks (3.0 Mo)	BD 4,570.909



Fixed Price Guarantee

Total project investment for Module 8 is fixed at **BD 4,570.909**. No hidden deployment, integration, or hourly surcharges apply within the agreed functional scope.

Detailed Milestone Breakdown & Deliverables (12-Week Roadmap)

The implementation is organized into 4 sequential 3-week milestones (25.00% each), guaranteeing focused execution, continuous integration, and transparent deliverable acceptance:

1

Milestone 1: Facility & Workplace Management, Work Orders & Utility Control

WEEKS 1 – 3 (3 WKS)

Setup of centralized Facility Master, preventive/corrective maintenance work orders, employee complaint ticketing, utility consumption logs, visitor registration (VMS), and commercial property lease agreement renewals.

✓ **Facility Master & Zones**

Centralized repository of all corporate branches, warehouses, and accommodation units.

✓ **Work Orders & Ticketing**

Maintenance calendar, AMC vendor contracts, and employee service request portal.

✓ **Utility & Lease Control**

Electricity/water cost tracking, budget variance, and automated lease renewal notifications.

2

Milestone 2: Fixed Asset Master, QR Tagging, Custodian Tracking & Depreciation

WEEKS 4 – 6 (3 WKS)

Engineering of Fixed Asset Master Registry, QR label generation, custodian assignments, inter-branch asset transfers, straight-line/reducing depreciation calculation, and physical audit verification tools.

3

✓ **QR Code Asset Tagging**

Unique QR label generation with handheld scanner barcode integration.

✓ **Asset Custody & Transfers**

Inter-branch transfer tracking, custodian sign-off, and movement history.

✓ **Depreciation & Audit Engine**

Automated monthly depreciation GL postings, warranty tracking, and scrap disposal.

Milestone 3: Fleet Logistics, Vehicle Bookings, Fuel Logs & TCO Analytics

WEEKS 7 – 9 (3 WKS)

Development of Vehicle Master Registry, driver allocation, GPS tracking integration, fuel consumption and card reconciliation, tire/parts service logs, insurance/road tax renewals, and total vehicle TCO analytics.

✓ **Vehicle Master & Bookings**

Fleet vehicle profiles, driver assignments, delivery trips, and key tracking.

✓ **Fuel & Mileage Analysis**

Fuel receipts, mileage efficiency (km/L), and fuel anomaly detection.

✓ **Compliance & TCO Dashboard**

Insurance/registration renewal alerts, maintenance logs, and vehicle TCO cockpit.

4

Milestone 4: Staff Locker Control, Secure Digital Document Vault & Final UAT

WEEKS 10 – 12 (3 WKS)

Implementation of Physical Locker Management, encrypted Digital Document Repository, physical document barcode movement tracking, automated statutory license renewal alerts (CR, Municipality), and multi-branch UAT sign-off.

✓ **Staff Locker Registry**

Branch locker allocation, key handovers, and locker inspection audits.



Secure Digital Document Vault

Encrypted repository for trade licenses, CRs, title deeds, with RBAC security.



Statutory Renewal Engine & UAT

Automated CR/permit renewal reminders, document movement tracking, and final UAT.

Milestone Payment Structure & Invoicing Schedule (12-Week Model)

Payments are structured in 4 equal milestone disbursements of **25.00% (BD 1,142.727 each)**, tied strictly to formal milestone deliverables and client sign-off:

MILESTONE PHASE	DELIVERABLE DESCRIPTION	TIMELINE	PERCENTAGE	AMOUNT (BHD)	PAYMENT TRIGGER & ACCEPTANCE CRITERIA
Milestone 1	Facility & Workplace Management, Work Orders & Utility Control	Weeks 1–3	25.00%	BD 1,142.727	Demonstration of Facility Master, Service Ticketing, Utility Logs, Lease Alerts
Milestone 2	Fixed Asset Master, QR Tagging, Custodian Tracking & Depreciation	Weeks 4–6	25.00%	BD 1,142.727	Sign-off of Asset Tagging, Scanner Integration, Transfers & Depreciation Engine
Milestone 3	Fleet Logistics, Vehicle Bookings, Fuel Logs & TCO Analytics	Weeks 7–9	25.00%	BD 1,142.727	Demonstration of Vehicle Master, Driver Bookings, Fuel Reconciliation & TCO Dashboard
Milestone 4	Staff Locker Control, Secure Digital	Weeks 10–12	25.00%	BD 1,142.727	Delivery of Locker Registry, Secure Digital

MILESTONE PHASE	DELIVERABLE DESCRIPTION	TIMELINE	PERCENTAGE	AMOUNT (BHD)	PAYMENT TRIGGER & ACCEPTANCE CRITERIA
	Document Vault & Final UAT				Digital Document Vault, CR Renewal Alerts & Fir Multi-Brand Sign-off
Total Milestone Project Cost	Complete Module 8 ERP Scope Delivery	12 Weeks	100.00%	BD 4,570.909	Formal Acceptance & Sign-off

Working Calendar, SLA, 15-Day Grace Period & Legal Terms

Working Calendar & Team Hours

Development operates on a 5-day standard work week (Sunday to Thursday, 09:00 to 18:00 AST), aligning with GCC financial business hours. Sprint demos are scheduled at the end of each 3-week milestone.

15-Day Milestone Review & Grace Period

Client is granted a formal 15-calendar-day review window following each sprint deployment to verify deliverables and audit tracking before milestone invoice closure.

SLA & Defect Severity Matrix

Critical Asset/Document Security Blockers (Severity 1) resolved within 4 hours; Major Functional Defects (Severity 2) within 24 hours; Minor UI adjustments (Severity 3) within 3 business days.

Scope Control & Document Security

Features adhere strictly to the 25-page specifications in ARCH-001. All legal document uploads and sensitive vehicle records are encrypted with AES-256 for corporate confidentiality.

Stakeholder Authorization & Sign-Off

By digitally endorsing this Milestone & Payment Structure document, all parties confirm adherence to the deliverables, 3-week timeline, and financial schedule defined for **SL-POP-ERP-MS-008**:

DEVELOPER / ARCHITECT

Ancy Varghese Thekkan
Software Architect, SaNDS Lab

✓ PREPARED (18-JUNE-2026)

COMPLETED

VERIFIED BY (DEVELOPER)

Ajit Kumar KV
CEO & Managing Director, SaNDS Lab

✓ VERIFIED (04-JULY-2026)

CHECKED & VERIFIED

CONSULTANT REVIEW

UniGlobal Business Solutions
Project Management Consultant

🕒 PENDING REVIEW

IN REVIEW

CLIENT APPROVAL

Popular Auto Spare & A/C Parts
Authorized Executive Signatory

🕒 PENDING APPROVAL

PENDING SIGNATURE